Beach Occupancy Monitoring for Coastal Resilience

Bao Nguyen (xnguyen2023@fau.edu), Dr. Soyoung Park, Dr. Jinwoo Jang
College of Engineering & Computer Science, College of Business

Abstract

This project proposes a computer vision system that estimate beach occupancy in real time from public live stream cameras, providing data to deepen understandings on beachgoer's behaviors and visitation patterns.

Background

Concentrated tourism at a small number of popular beaches contributes to:
- Increase in litter and waste
- Strain on local safety and infrastructure resources
- Reduction in quality of life for residents

Meanwhile, nearby beaches receive little visitor traffic, having less opportunities for tourist-driven economic activity and growth.

This project aims to provide data regarding beach visitor's patterns and behaviors. These patterns and behaviors can help inform strategies to better distribute beachgoers across different locations.

Data Sources

- Boca Raton Beach Webcam (Erdman Video Systems, Inc., n.d.)
- Miami and Beaches Webcams (Greater Miami Convention & Visitors Bureau, n.d.)

Methodology

The system pipeline consists of four stages:
1. Capture: Periodically pull still frames from live HLS beach webcam streams using OpenCV.
2. Baseline detection: Apply a YOLOv8 object detection model (Jocher et al., 2023) to count visible individuals per frame.
3. Image Annotation: Overlay the detected people count on the image and save it to an output folder.
4. Data Logging: Record each frame's timestamp and detected people count to a CSV file for later time-series analysis.

Preliminary Results

A proof-of-concept was built and tested on a live beach stream capture:

- Frame capture pipeline successfully pulled and saved timestamped frames from a live HLS stream

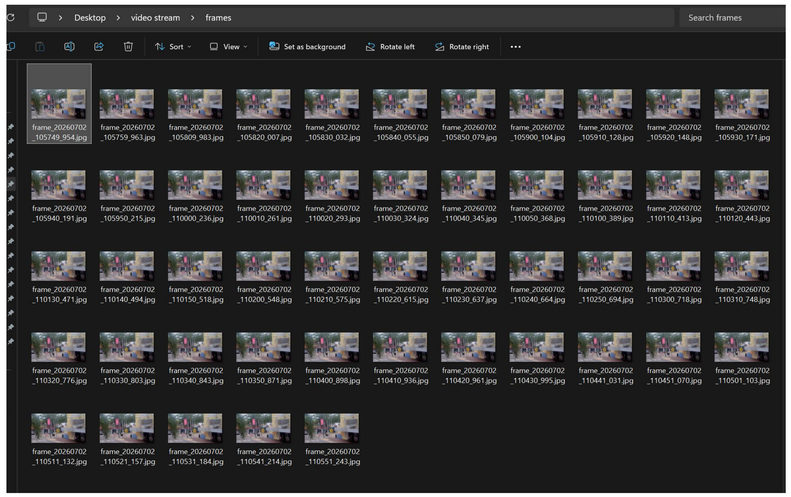


Figure 1: Video frames are successfully pulled from public live stream

- YOLOv8 baseline detected 12 people in an initial test frame out of 19 people present on the beach.

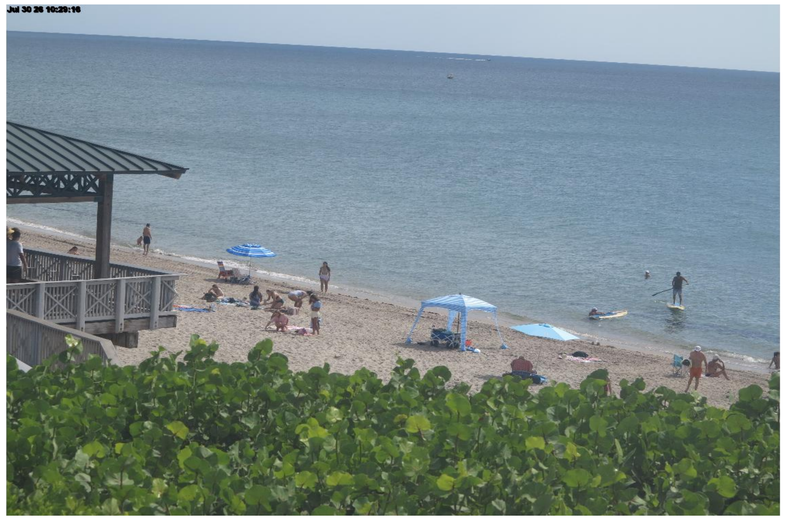


Figure 2: Unprocessed video frame pulled from public live stream

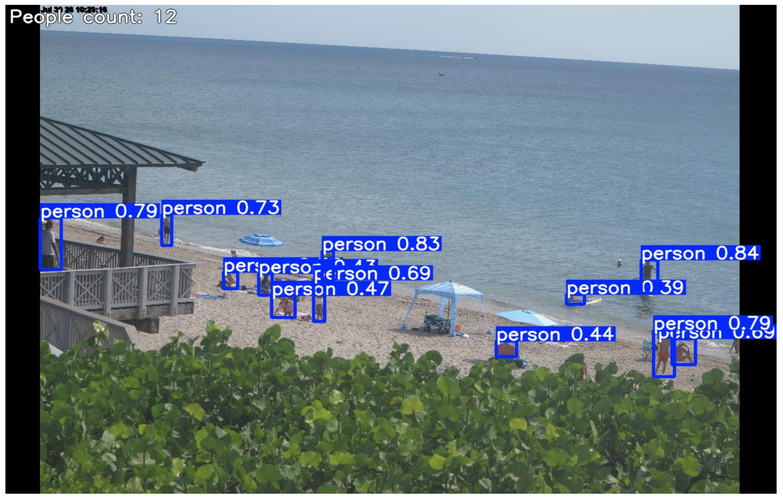


Figure 3: Annotated video frame

Implication

These preliminary results show that an automated beach occupancy estimation system is technically feasible, establishing the foundation for future occupancy comparison between beaches and analysis of visiting trends over time.

Next Steps

- Implement a pretrained density-map crowd-counting model
- Quantitatively compare density-map vs. object-detection accuracy, especially under occlusion
- Scale the pipeline to multiple beach camera feeds for cross-beach comparison
- Extract behavioral patterns from the image to help guide redistribution strategies.

References

1. Erdman Video Systems, Inc. (n.d.). Boca Raton beach webcam. https://video-monitoring.com/beachcams/boca/
2. Greater Miami Convention & Visitors Bureau. (n.d.). Miami webcams. Miami and Beaches. https://www.miamiandbeaches.com/plan-your-trip/miami-webcams
3. Jocher, G., Chaurasia, A., & Qiu, J. (2023). Ultralytics YOLOv8 [Computer software]. https://github.com/ultralytics/ultralytics

This material is based upon work supported by the U.S. Department of Education under Research and Development Infrastructure Grant No. P116H230018. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the U.S. Department of Education.